

TECHNICAL & LEGAL COMPLIANCE WHITEPAPER

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FGF GUILD MANAGEMENT

Technical Architecture, Zero-Trust Security, Privacy by Design & Intellectual Property Compliance Whitepaper

Formal Purpose of this Document

This official whitepaper is prepared specifically for the **Game Publisher, Developer, Legal Counsel, and Community Operations teams**. It demonstrates unequivocally that the FGF Guild Management Platform is an independent, non-intrusive community companion tool that processes **Zero Personal Identifiers (Zero PII)**, incorporates **Zero Copyrighted Game Assets**, does **not interact with game client memory or network streams**, and enforces enterprise-grade **PostgreSQL 17 Row-Level Security**.

0

PERSONAL DATA (ZERO PII)

0

COPYRIGHTED GAME
ASSETS

0

GAME MEMORY / CLIENT
HOOKS

100%

POSTGRESQL RLS
PROTECTED

TARGET AUDIENCE

Game Publisher & Developer Compliance / Security Teams

DOCUMENT VERSION

2.4.0 (Consolidated Canonical Specification)

PRIMARY FRAMEWORKS

GDPR, CCPA, OWASP Top 10, Zero-Trust Architecture

DATE OF ATTESTATION

August 19, 2026

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1. Executive Summary & Scope

1.1 Mission & Nature of the Platform

The **FGF Guild Management Platform** is an out-of-game, web-based software-as-a-service (SaaS) companion dashboard designed solely to assist guild officers and players in organizing internal rosters, scheduling guild battle events, tracking aggregate statistical participation, and dispatching announcements via Discord webhooks.

The platform operates **100% externally and asynchronously** from the video game client and its operational servers. It acts in the exact same legal and technical capacity as a collaborative spreadsheet (e.g. Google Sheets or Microsoft Excel) or an out-of-game Discord bot, providing community organization tools.

1.2 Fundamental Compliance Invariants

 **Game Integrity & Non-Interference**

- ✗ **Zero Client Hooks:** No injection into game memory or executable space.
- ✗ **Zero Network Tampering:** No packet interception or sniffing.
- ✗ **Zero Automation:** No automated botting, macros, or gameplay actions.
- ✗ **Zero Private APIs:** No unauthorized reverse-engineering of game endpoints.

 **Data Privacy & IP Safeguards**

- ✓ **Zero PII Collected:** No real names, emails, phones, or GPS coordinates.
- ✓ **Zero Game Assets:** No copyrighted textures, sprites, or 3D models.
- ✓ **Ephemeral In-Memory OCR:** Roster screenshots destroyed immediately after parse.
- ✓ **PostgreSQL 17 RLS:** Kernel-level cryptographic tenant isolation.

2. Zero-PII & Privacy-by-Design Governance (GDPR & CCPA)

The platform was architected under strict **Privacy by Design** (PbD) and **Data Minimization** principles in full alignment with the European General Data Protection Regulation (GDPR - Regulation EU 2016/679), the California Consumer Privacy Act (CCPA), and child safety standards (COPPA).

2.1 Strict Absence of Personally Identifiable Information (PII)

The database contains **zero real-world personal data**. Below is an exhaustive audit of all data fields handled by the system:

DATA CATEGORY	STORED?	TECHNICAL IMPLEMENTATION & PRIVACY SAFEGUARD
Real Names / Surnames	NO (0%)	Only public in-game nicknames chosen by players are stored (e.g. "StarLord99"). No legal names are ever requested or accepted.
Personal Email Addresses	NO (0%)	Application database tables store zero real emails. Player accounts use internal synthetic tokens or pseudonymized login handles. Supabase Auth identity is strictly isolated.
Physical Address / GPS	NO (0%)	Zero geographical tracking or GPS coordinates exist. No device geolocation APIs are invoked.
Phone / SMS Numbers	NO (0%)	No telephony or SMS integrations exist.
Financial / Card Details	NO (0%)	SaaS subscriptions are handled 100% off-site by Stripe Hosted Checkout (PCI-DSS Level 1). Zero credit card numbers or billing addresses touch our database.
In-Game Passwords / Auth	NO (0%)	The platform never prompts for, accepts, or stores video game passwords, Google Play credentials, Apple IDs, or game session tokens.
IP Address Persisted	NO (0%)	IP addresses are never persisted in database tables. Cloud infrastructure logs undergo automated short-term rotation.
Timezone Offset	Offset Only	Only a non-identifying integer offset (e.g. +2 or -5) is stored to calculate guild battle countdowns relative to UTC.

Data Minimization & Cascade Deletion

All member records maintain strict foreign key cascades (`ON DELETE CASCADE`). When an officer or member removes their profile, all related event participations, squad assignments, and score logs are **instantaneously and permanently deleted** from the database.

Zero Ad Trackers & Behavioral Telemetry

The client contains **zero marketing SDKs**, zero advertising cookies, and zero behavioral telemetry tools (no Facebook Pixel, no Google Ads, no data brokering). Data is exclusively used for guild roster display.

3. Intellectual Property (IP), Asset & Trademark Safeguards

A foundational design requirement is the absolute respect of the Game Publisher’s copyright, trademarks, game assets, and intellectual property.

3.1 100% Zero Copyrighted Game Asset Guarantee

The application repository and production distribution bundles have been verified to contain **zero copyrighted game assets**:

- ✗ **Zero Game Textures / Bitmaps**: No spaceship textures, character portraits, weapon icons, or UI artwork from the video game are hosted or served.
- ✗ **Zero 3D Models / Meshes**: No proprietary 3D geometry, polygon meshes, or skeletal animations are used.
- ✗ **Zero Audio Tracks / Effects**: No in-game sound effects, voice lines, or musical scores are distributed.
- ✗ **Zero Game Code Binaries**: No decompiled game code, reverse-engineered bytecode, or shaders are included.

3.2 Permissive Open-Source UI Design System

All visual components and interfaces are constructed exclusively from open-source, permissively licensed design libraries:

INTERFACE LAYER	DESIGN STANDARD & PROVENANCE	LICENSE TYPE
UI Framework & Tokens	Google Material Design 3.0 (M3) Hierarchy styled with Cozy Graphite / Titanium neutral surfaces	Open Standard / MIT
Typography	Google Fonts <i>Plus Jakarta Sans</i> & <i>Inter</i>	SIL Open Font License 1.1
Iconography	Google Material Symbols Rounded (Variable Font Glyphs)	Apache License 2.0
3D Ambient Background	Procedural, mathematically generated geometric particle scene using Three.js	MIT License
Application Charts	Custom HTML5 2D Canvas vector rendering (<code>PortalChart.ts</code>)	Proprietary Original Code

3.3 Ephemeral In-Memory OCR Processing (Zero Screenshot Storage)

To assist guild officers with roster management without manual typing, an Optical Character Recognition (OCR) pipeline is provided. The architecture guarantees that **no user screenshots are permanently saved or stored**:

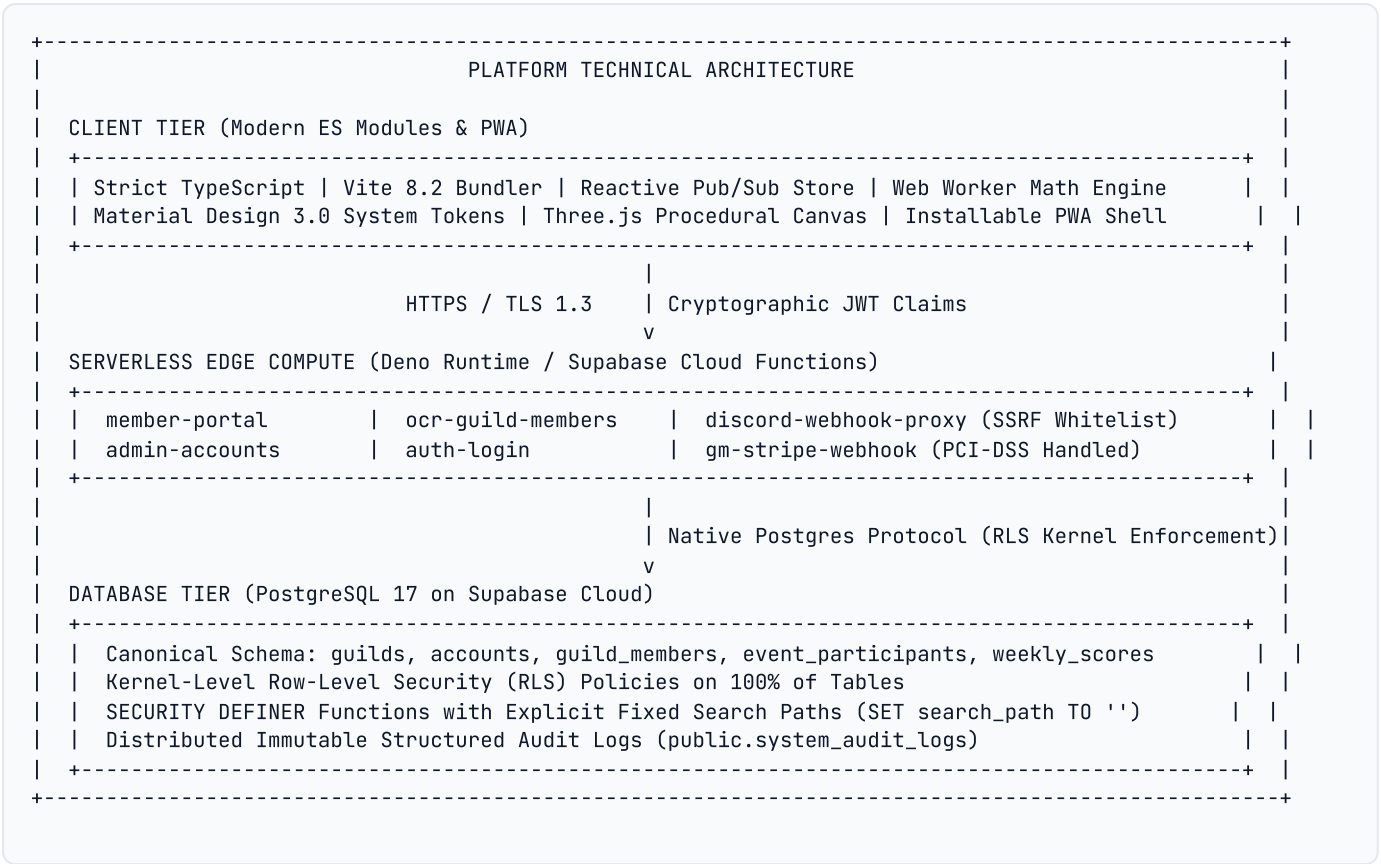
- In-Memory Transit**: The guild officer uploads a leaderboard screenshot over an encrypted TLS 1.3 channel to the `ocr-guild-members` Edge Function.
- AI Vision Extraction**: Google Gemini Vision API parses the image in volatile RAM to extract numeric values (power, scores) and public pseudonyms.
- Immediate Bitmap Destruction**: The screenshot bitmap is **immediately destroyed from RAM** upon parse completion. The platform maintains zero disk storage, zero S3/bucket image retention, and zero image archiving.
- Defensive Sanitization**: Extracted values are strictly sanitized into plain integers before insertion into database tables.

Explicit Legal Disclaimer of Non-Affiliation

"FGF Guild Management is an independent, community-created companion tool. It is not affiliated with, endorsed by, sponsored by, or specifically approved by the game developer or publisher. All game titles, trademarks, and associated names remain the exclusive intellectual property of their respective owners."

4. Technical Architecture & Modern Serverless Stack

The platform employs a decoupled, modern serverless architecture built on modern web standards, strict static type safety, and managed cloud infrastructure.



⚡ Client Tier

- Vite 8.2 & TypeScript 7
- Reactive Pub/Sub Store
- Web Workers for calculations
- Zero heavy UI frameworks

🌐 Edge Compute

- Deno TypeScript Runtime
- Global low-latency deployment
- SSRF-protected proxies
- Zero server maintenance

🗄️ Database Tier

- PostgreSQL 17 Cloud
- 100% RLS Coverage
- AES-256 Encryption at rest
- Immutable Audit Logs

4.1 Distributed Observability & Structured Audit Logging

Every edge execution and browser transaction generates structured JSON telemetry with correlation tracking:

- Correlation IDs (`x-correlation-id`):** Distributed end-to-end request tracing across client and edge functions.
- Automated Credential Scrubbing:** Passwords, tokens, and authorization headers are automatically redacted prior to logging.
- Persistent Audit Trail:** Administrative mutations (roster updates, member role promotions) write permanent records to `public.system_audit_logs`.

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5. Zero-Trust Security Model & Defensive Protections

The platform enforces a strict **Zero-Trust Security Model**: all requests require explicit cryptographic verification regardless of origin.

5.1 The Four-Role Access Control Model (RBAC)

ROLE	DATABASE & API SCOPE	SECURITY ISOLATION BOUNDARY
<code>super_admin</code>	Platform owner. Reads & writes all guild tenants for maintenance and global diagnostic logs.	Single master account (HawkEye). Enforced via JWT verification + database role checks.
<code>server_admin</code>	Reads & writes only guild tenants sharing their assigned game <code>server_number</code> .	PostgreSQL RLS queries cross-checking <code>guilds.server_number</code> . Cross-server writes are blocked.
<code>guild_admin</code>	Reads & writes strictly their single assigned guild tenant (e.g. <code>ALPHA</code>).	PostgreSQL RLS engine isolation. Attempting to access another guild returns empty sets or 403.
<code>member</code>	ZERO direct database access . Cannot read or write any database tables directly.	Restricted to Player Portal. Communicates solely via the <code>member-portal</code> Edge Function.

5.2 PostgreSQL 17 Kernel Row-Level Security (RLS)

Tenant isolation is enforced directly within the PostgreSQL kernel via `SECURITY DEFINER` access functions with explicit `SET search_path TO ''` to prevent search path hijacking:

```
-- Centralized Tenant Read Qualifier Function
CREATE OR REPLACE FUNCTION public.gm_can_read_guild_data(p_guild text)
RETURNS boolean LANGUAGE sql STABLE SECURITY DEFINER SET search_path TO ''
AS $$
    SELECT EXISTS (
        SELECT 1 FROM public.accounts a
        LEFT JOIN public.guilds g ON g.id = p_guild
        WHERE a.auth_user_id = (SELECT auth.uid())
              AND a.status = 'active'
              AND (
                  a.role = 'super_admin'
                  OR (a.role = 'server_admin' AND a.server_number IS NOT NULL AND a.server_number =
g.server_number)
                  OR (a.role IN ('guild_admin', 'R5', 'R4', 'admin') AND a.guild = p_guild)
              )
    );
$;
```

5.3 Defensive Edge Function Hardening

SSRF Webhook Whitelisting

The `discord-webhook-proxy` enforces strict hostname whitelisting (`discord.com/api/webhooks/`). Arbitrary destination URLs, loopback interfaces (`127.0.0.1`), and internal RFC 1918 private subnets are categorically blocked.

Defensive Score Bounding

All numerical submissions pass through `parseSafeScore` , bounding non-negative integer values to `500,000,000` . This prevents integer overflow attacks, NaN corruption, and database schema errors.

6. Automated Quality Assurance & Verification Protocols

The platform’s stability and security invariants are guaranteed via an automated continuous integration (CI) pipeline on GitHub Actions:

Continuous Integration Quality Gate (.github/workflows/ci.yml)

Step 1

Type Check

tsc --noEmit (0 errors)

Step 2

Unit Tests

Vitest (219/219 green)

Step 3

Prod Build

Vite bundle compile

Step 4

E2E Testing

Playwright tests

7. Publisher Compliance Matrix & Summary Checklist

COMPLIANCE DOMAIN	PUBLISHER REQUIREMENT	PLATFORM SAFEGUARD & ARCHITECTURE	STATUS
Personal Data (PII)	Zero real-world personal identification stored.	Only public in-game nicknames and non-identifying stats stored.	VERIFIED
Game Client Memory	No hooks, DLL injection, or memory inspection.	100% external web browser application. Zero memory interaction.	VERIFIED
Game Automation / Bots	No gameplay macros, farming bots, or automated actions.	Manual roster coordination tool. Zero game automation.	VERIFIED
Copyrighted Assets	No proprietary game textures, audio, or 3D models.	100% open-source UI (Material Design 3.0, Material Symbols, Three.js).	VERIFIED
OCR Screenshot Policy	Screenshots must not be archived or monetized.	In-memory OCR extraction. Bitmaps destroyed immediately from RAM.	VERIFIED
Access Security & RLS	Robust multi-tenant separation.	PostgreSQL 17 Row-Level Security on 100% of tables.	VERIFIED
Network Safety (SSRF)	Safe webhook proxies without internal attack vectors.	Strict Discord domain whitelist; blocks private subnets.	VERIFIED

8. Official Attestation & Responsible Communication

Formal Platform Attestation

This document certifies that the **FGF Guild Management Platform** operates exclusively as a community-driven organizational companion. It poses zero threat to the Game Publisher's server infrastructure, game economy, copyright assets, player privacy, or intellectual property rights.

Contact & Publisher Inquiries

The development team welcomes active dialogue with the Game Publisher and its community safety representatives. For technical questions, compliance verifications, or security coordination, please reach out through the official repository channels.